

K-Nearest Neighbours

Parametric Models : Parametric Models are those where the hypothesis function h can be indexed by a fixed number of parameters.

That is, the number of parameters of the model (hypothesis) are fixed, and do not change with the increase ~~with~~ in the data set size.

↳ Linear Regression

↳ Logistic Regression

↳ Neural Networks

Non-Parametric Methods : The number of parameters describing the model/hypothesis usually grows with the dataset size.

↳ In Decision Trees, the parameters of the model, which is the tree size, is certainly not fixed and usually changes with the size of the data.

For example, smaller datasets could be modeled with small size trees and larger datasets could require bigger trees.

- KNN

- Classification & Regression

- Decision Trees

- Classification & Regression

These models are more flexible to the shape of the training data, but this sometimes comes at the cost of interpretability.

KNN Idea

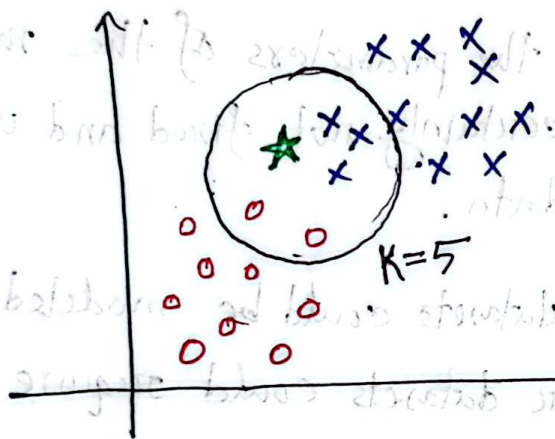
Step-1 Load training data

Step-2 Find out the similarity/distance of ~~test~~ test examples with all points in the training data.

Step-3 Sort the results according to distance/similarity

Step-4 Take mean of nearest K values for Regression,
or take majority label from nearest K values for classification.

Classification



3 - blue

2 - Red

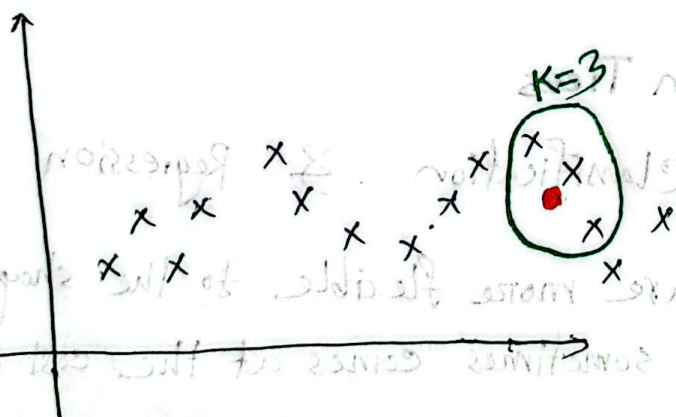
$$\therefore \text{Prediction (Blue)} = \frac{3}{5} \times 100\%$$

$$= 60\%$$

$$\text{Prediction (Red)} = \frac{2}{5} \times 100\%$$

$$= 40\%$$

Regression



House price

Avg of 3 house.